

# Bill of Materials — t8\_r3 (VBF-T8R3-BOM-01)

## Sheet: BOM

| Component                              | Mass (g/cell) | kg/kWh      | Basis / volume fraction               | Source / annotation                                         |
|----------------------------------------|---------------|-------------|---------------------------------------|-------------------------------------------------------------|
| Positive electrode active material     | 16.1685       | 0.8708      | 96 wt% of positive layer              | literature default (parameter set lacks AM/CB/binder split) |
| Positive electrode conductive additive | 0.3368        | 0.0181      | 2 wt%                                 | literature default                                          |
| Positive electrode binder (PVDF)       | 0.3368        | 0.0181      | 2 wt%                                 | literature default                                          |
| Negative electrode active material     | 9.7432        | 0.5247      | 96 wt% of negative layer              | literature default                                          |
| Negative electrode conductive additive | 0.1015        | 0.0055      | 1 wt%                                 | literature default                                          |
| Negative electrode binder (CMC/SBR)    | 0.3045        | 0.0164      | 3 wt%                                 | literature default                                          |
| Separator                              | 0.1945        | 0.0105      | 9 $\mu\text{m}$ , porosity 0.47       | parameter set + energy:layer_kg_m2                          |
| Electrolyte                            | 6.7925        | 0.3658      | pore volume 5.66 $\text{cm}^3$        | 1.2 $\text{g/cm}^3$ literature value $\times$ fill 1.0      |
| Positive current collector (Al)        | 2.7729        | 0.1493      | 10 $\mu\text{m}$                      | round-1 override; energy:layer_kg_m2                        |
| Negative current collector (Cu)        | 7.3615        | 0.3965      | 8 $\mu\text{m}$                       | round-1 override; energy:layer_kg_m2                        |
| Enclosure / tabs                       | Not modeled   | Not modeled | —                                     | beyond pure simulation boundary                             |
| TOTAL (incl. electrolyte)              | 44.1127       | 2.3757      | —                                     | sum of rows above                                           |
| Stack total (contract caliber)         | 37.3203       | 2.0099      | $\Sigma$ layers, electrolyte excluded | energy:mass_kg — the 37.32 g used for ED                    |

## Sheet: Summary

| Metric                                     | Value               | Source                      |
|--------------------------------------------|---------------------|-----------------------------|
| Cell energy (kWh)                          | 0.01856818981379996 | energy:energy_wh/1000       |
| Total mass incl. electrolyte (kg)          | 0.04411274722660002 | BOM sheet                   |
| Material usage (kg/kWh, incl. electrolyte) | 2.375716085895203   | BOM sheet                   |
| Stack mass (kg, contract)                  | 0.03732025138660001 | energy:mass_kg              |
| Energy density (Wh/kg, contract)           | 497.5365685898607   | energy:energy_density_wh_kg |